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Discrete Tomography

Bachelor Project

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Contents

Introduction

Discrete tomography is a branch of computational imaging that focuses on reconstructing discrete images from a limited set of projections. Unlike classical tomography, where continuous images are reconstructed from dense projection data, discrete tomography deals with images composed of a finite number of intensity levels, often binary or small integer values. This restriction arises naturally in applications such as medical imaging, materials science, and industrial inspection, where objects are composed of a small set of distinguishable components.

Chapter 1 brings an overview of different types of encodings used to transform high-level constraints into conjunctive normal form (CNF), since SAT solvers operate exclusively on CNF formulas. Therefore, we present several commonly used encoding techniques, discuss their principles, and emphasize their strengths and weaknesses.

Efficient representation and encoding of images are crucial topics in discrete tomography, as the reconstruction process depends heavily on how projection data are stored and processed. Chapter 2 introduces several projecting strategies, including standard rows, columns, and diagonal projections, but also slope-diagonal and frame-based ones. Once we have the set of projections of the image, the reconstruction problem can be encoded into a set of logical constraints, which can then be solved using a SAT solver.

Discrete tomography raises fundamental theoretical questions about the uniqueness and number of possible solutions for a given set of projections. Chapter 3 addresses estimating upper bounds on the number of solutions and demonstrates how the solution space scales with the image size. Overall, theoretical analysis provides a comprehensive perspective on discrete tomography as a computational discipline.

Chapter 1

SAT Encodings of Cardinality Constraints

Cardinality constraints are a fundamental class of constraints in Boolean satisfiability problems. They restrict the number of variables that may be assigned true value within a given set, typically enforcing conditions of the form “at most k”, “at least k”, or “exactly k” true variables. Such constraints naturally arise in a wide range of applications, including combinatorial reconstruction problems such as discrete tomography.

Since SAT solvers operate on formulas in conjunctive normal form (CNF), cardinality constraints must be translated into CNF using dedicated encoding techniques. A naïve translation by enumerating all forbidden combinations is generally infeasible, as it leads to an exponential grow in the number of clauses. Consequently, a variety of specialized encodings (such as sequential counter encoding, totalizer encoding, and cardinality networks) have been proposed to represent cardinality constraints compactly and efficiently. These encodings differ in their structural properties, the number of auxiliary variables, and clauses they introduce.

In this chapter, we present an overview of these encoding techniques and discuss their theoretical characteristics. We define them based on Donald Knuth’s book [1], and describe their suitability for encoding constraints arising in discrete tomography problems, where large numbers of cardinality constraints are generated over overlapping variable sets.

1.1 Sequential Counter Encoding

The Sequential Counter Encoding, introduced by Carsten Sinz around 2005, is one of the most important and widely used in SAT. [2] Let $x_1, \ldots, x_n$ be Boolean variables and let $r \in \mathbb{N}$. Sinz's method introduces $(n-r)r$ auxiliary variables

$$s_j^k \quad \text{for } 1 \le j \le n-r,\ 1 \le k \le r.$$

These variables have the following interpretation: $s_j^k = 1$ means that among $x_1, \ldots, x_{j+k-1}$ there are at least k true variables. The cardinality constraint

$$x_1 + \cdots + x_n \le r$$

is encoded by adding the following clauses:

$$(\neg s_j^k \vee s_{j+1}^k), \quad 1 \le j \le n-r-1,\ 1 \le k \le r, \tag{1.1}$$

$$(\neg x_{j+k} \vee \neg s_j^{k-1} \vee s_j^k), \quad 1 \le j \le n-r,\ 1 \le k \le r. \tag{1.2}$$

Here, s_j^0 is assumed to be true and s_j^{r+1} is omitted.

If F is any satisfiability problem and if we add the $(n-r-1)r+(n-r)(r+1)$ clauses of the forms (1.1) and (1.2), then the new set of clauses is satisfiable if and only if F is satisfiable with $x_1 + \cdots + x_n \le r$.

1.2 Totalizer Encoding

Another way to achieve the same goal, which has been proposed by O. Bailleux and Y. Boufkhad around 2003, is called totalizer encoding. [3] Consider a complete binary tree that has $n-1$ internal nodes numbered 1 through $n-1$, and n leaves numbered n through $2n-1$; the children of node k, for $1 \le k < n$, are nodes $2k$ and $2k+1$. We form new variables b_i^k for $1 \le k < n$ and $1 \le j \le t_k$, where t_k is the minimum of r and the number of leaves below node k. When $b_i^k = 1$, it means that among the leaves in the subtree rooted at node k, at least j of them are true. Then the following clauses do the job:

$$(\bar{b}_i^{2k} \vee \bar{b}_{i+1}^{2k+1} \vee b_{i+j}^k), \quad 0 \le i \le t_{2k},\ 0 \le j \le t_{2k+1},\ 1 \le i+j \le t_k+1,\ 1 < k < n; \tag{1.3}$$

$$(\bar{b}_i^2 \vee \bar{b}_j^3), \quad 0 \le i \le t_2,\ 0 \le j \le t_3,\ i+j = r+1. \tag{1.4}$$

In these formulas we let $t_k = 1$ and $b_1^k = x_{k-n+1}$ for $n \le k < 2n$; all literals $\bar{b}_0^k$ and b_{r+1}^k are to be omitted. Compared to simpler encodings, totalizer encoding offers a good trade-off between the number of auxiliary variables and clauses, while also offering strong propagation properties in SAT solvers. In particular, it supports efficient unit propagation for upper-bound constraints, making it suitable for practical SAT-based solving.

1.3 Cardinality Networks

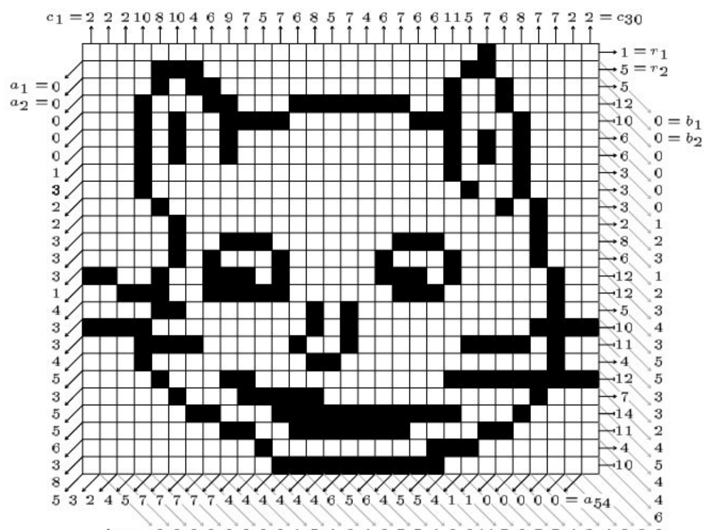
Another widely used approach for encoding cardinality constraints is based on cardinality networks. These encodings rely on sorting networks, such as bitonic sorting networks, to sort the input Boolean variables according to their truth values. The sorted outputs then represent the number of variables set to true, and a constraint of the form $\sum_{i=1}^{n} x_i \leq k$ can be enforced by fixing the $(k+1)$-st output of the network to false. [4]

Cardinality network encodings are known for their strong propagation properties, which often lead to better solver performance compared to simpler encodings. However, this strength comes at the cost of a larger number of auxiliary variables and clauses, typically $\Theta(n \log^2 k)$, which can make the encoding less space-efficient than the alternatives.

Chapter 2

Image Description & Runtime Analysis

In the following paragraphs, we are going to focus on the study of binary images for which partial information is given. We will see how the amount of information given influences the running time and the number of solutions. As a reference example, we are going to use a “Cheshire cat” which is a bitmap of $m = 25$ rows and $n = 30$ columns.



Cheshire cat — an array of black and white pixels together with its row sums r_i, column sums c_j, and diagonal sums a_d, b_d.
Adapted from [1, p. 25].

2.1 Standard Description

Let $X = (x_{i,j}) \in \{0,1\}^{m\times n}$ be an $m \times n$ bitmap. The standard encoding consists of row, column, and diagonal projections. We are given the row sums $r_i = \sum_{j=1}^{n} x_{i,j}$ for $1 \leq i \leq m$ and the column sums $c_j = \sum_{i=1}^{m} x_{i,j}$ for $1 \leq j \leq n$, as well as both sets of sums in the 45° diagonal direction, namely $a_d = \sum_{i+j=d+1} x_{i,j}$ and $b_d = \sum_{i-j=d-n} x_{i,j}$ for $0 < d < m+n$. We will refer to these special diagonals as the regular diagonals.

These vectors, which contain the sums, can be encoded as a sequence of clauses to be satisfied using one of the encodings described earlier. The following tables compare these encodings using the SAT solver CaDiCaL 1.9.5. For each encoding, we report the number of variables and clauses, as well as the build time and solve time. Build time refers to the time required to construct the CNF encoding in memory, while solve time measures the time the SAT solver takes to find a solution or determine unsatisfiability. All computations were performed using the PySAT library.

Scenario	Encoding	# vars	# clauses	Build (s)	Solve (s)
r_i, c_j	seqcounter	14833	28195	0.1289	0.0721
	totalizer	42497	62676	0.2704	0.1520
	cardnetw	36704	53987	0.1884	0.1157
a_d, b_d	seqcounter	10125	18888	0.0742	0.0379
	totalizer	42655	63039	0.1992	0.3652
	cardnetw	33281	48982	0.1998	0.1783
r_i, c_j, a_d, b_d	seqcounter	24208	47083	0.3207	9.3721
	totalizer	84402	125715	0.4066	60.5126
	cardnetw	69235	102969	0.5509	51.9776

Table 2.1: Comparison of SAT encodings across standard scenarios

2.2 Slope-Diagonal Description

First, we have to carefully define slope diagonals. Let $k \in \mathbb{N}$ be a fixed slope parameter. For $d = 1, \ldots, m + \lceil \frac{n}{k} \rceil - 1$, we define the *A-slope diagonal sums* by

$$a_d^{(k)} = \sum_{\substack{1 \le i \le m \\ 1 \le j \le n \\ i + \lfloor \frac{j-1}{k} \rfloor = d}} x_{i,j}.$$

Similarly, we define the *B-slope diagonal sums* by

$$b_d^{(k)} = \sum_{\substack{1 \le i \le m \\ 1 \le j \le n \\ (m-i+1) + \lfloor \frac{j-1}{k} \rfloor = d}} x_{i,j}.$$

Intuition: The parameter k determines the slope of the diagonals. For A-slope diagonals, k corresponds to k steps to the right and 1 step up. For B-slope (anti-) diagonals, it corresponds to k steps to the right and 1 step down. Note that for $k = 1$ we recover the regular diagonal sum vectors, in other words, $a_d = a_d^{(1)}$ and $b_d = b_d^{(1)}$ for every $d = 1, \ldots, m + \lceil \frac{n}{k} \rceil - 1$.

Suppose that we are given $a_d^{(k)}$ and $b_d^{(k)}$ for a fixed k. In the following tables, we can see how $a_d^{(k)}$ and $b_d^{(k)}$ together with row and column vectors influence the running time for various slopes k.

Scenario	Encoding	# variables	# clauses	Build time (s)	Solve time (s)
$a_d^{(2)}, b_d^{(2)}$	seqcounter	13009	24621	0.0841	0.0442
	totalizer	39648	58491	0.1724	0.2329
	cardnetw	33405	49130	0.1664	0.1838
$a_d^{(5)}, b_d^{(5)}$	seqcounter	15235	29027	0.0922	0.0463
	totalizer	38915	57334	0.1886	0.1048
	cardnetw	34038	50021	0.2022	0.0942
$a_d^{(10)}, b_d^{(10)}$	seqcounter	15332	29202	0.1085	0.0542
	totalizer	39304	57904	0.2020	0.1198
	cardnetw	34116	50124	0.1958	0.1112
$r_i, c_j, a_d^{(2)}, b_d^{(2)}$	seqcounter	27092	52816	0.2423	13.1813
	totalizer	81395	121167	0.3809	16.3476
	cardnetw	69359	103117	0.4602	18.6142
$r_i, c_j, a_d^{(5)}, b_d^{(5)}$	seqcounter	29318	57222	0.2010	4.5283
	totalizer	80662	120010	0.4405	19.5546
	cardnetw	69992	104008	0.4139	26.8208
$r_i, c_j, a_d^{(10)}, b_d^{(10)}$	seqcounter	29415	57387	0.2094	3.4116
	totalizer	81051	120580	0.4849	47.9697
	cardnetw	70070	104111	0.4053	21.2280

Table 2.2: Comparison of SAT encodings for different slope diagonals constraints

2.3 Frame-Based Description

For

$$s = 1, \ldots, \left\lceil \frac{\min(m,n)}{2} \right\rceil,$$

we define the *s-th frame sum* by

$$f_s = \sum_{\substack{s \leq i \leq m-s+1 \\ s \leq j \leq n-s+1 \\ (i=s) \vee (i=m-s+1) \vee (j=s) \vee (j=n-s+1)}} x_{i,j}.$$

The results below demonstrate what happens if we create new constraints from given sums f_s. As we have already seen in the previous section, sequential counter encoding renders best results.

Scenario	Encoding	# variables	# clauses	Build time (s)	Solve time (s)
f_s	seqcounter	19221	36947	0.1434	0.0701
	totalizer	46333	68388	0.2546	0.1277
	cardnetw	35648	52361	0.1880	0.1021
r_i, c_j, f_s	seqcounter	33304	65142	0.2586	2.5813
	totalizer	88080	131064	0.4846	35.0899
	cardnetw	71692	106348	0.4174	8.8998

Table 2.3: Comparison of SAT encodings for frame constraints

2.4 Important Remarks

All of the reported runtime measurements were obtained through local testing on a personal computer. To draw reliable and statistically meaningful conclusions, it would be necessary to run these experiments on a server or, for example, on an RCI cluster.

Furthermore, it is possible to combine these descriptions and use more of them to encode an image. That would lead to a decrease in the number of solutions but also to a big increase in the running time. Unfortunately, I was unable to get any reasonable results on my computer.

Chapter 3

Theoretical Analysis of the Number of Solutions

In this chapter, we are going to try to estimate a number of solutions for a given set of projections of an image. Because it is computationally demanding to determine the exact number of solutions for larger images, we will provide some theoretical arguments and bound the number of solutions from above, and also show that solutions grow with the size of the image.

3.1 Estimating the Upper Bound on Solutions

Our goal is to determine a generous upper bound on the possible number of different sets of input data $\{r_i, c_j, a_d, b_d\}$ that might be given to an $m \times n$ digital tomography problem, assuming that each of these sums independently has any of its possible values. We will also show how this bound compares to 2^{mn}, which is the total number of bitmaps of size $m \times n$. We will demonstrate our result on a bitmap with $m = 25$, $n = 30$.

We start with some basic observations. Since the length of the vectors is bounded, we obtain the following inequalities:

$$0 \leq r_i \leq n$$

$$0 \leq c_j \leq m$$

$$0 \leq a_d, b_d \leq min\{d, m, n, m + n - d\}$$

Without loss of generality, suppose that $m \leq n$. Each of the m rows consists of n pixels, so the sum of each row can take values in $\{0, 1, \ldots, n\}$. Thus, there are $(n + 1)$ possible values for each row sum, and consequently $(n + 1)^m$ possible distinct row sum vectors in total. Using the same reasoning, we obtain that there are $(m+1)^n$ possible distinct column sum vectors in total.

Next, note that there are exactly $m+n-1$ diagonals and also anti-diagonals. Thus, we only have to compute the number of possibilities for diagonals and then simply square it. A diagonal with index d contains exactly $min\{d, m, n, m+n-d\}$ pixels. Thus, the total number of possible distinct diagonal sum vectors is

$$\prod_{d=1}^{m+n-1} (1 + min\{d, m, n, m+n-d\})$$

We can further simplify this expression knowing that the lengths are $1, 2, \ldots, m, m, \ldots, m, m-1, \ldots, 2, 1$. Thus, the product becomes

$$\prod_{k=1}^{m} (k+1) \cdot (m+1)^{n-m} \cdot \prod_{k=1}^{m-1} (k+1)$$

Using factorials, we can express the number of possibilities as

$$(m+1)! \cdot (m+1)^{n-m} \cdot m!$$

Putting it all together, we have shown that the total number $B(m,n)$ of possible combinations of row, column, and diagonal sums is

$$(n+1)^m (m+1)^n ((m+1)!(m+1)^{n-m} m!)^2$$

One can compute that $B(25, 30) \approx 3 \times 10^{197}$. If we roughly assume that bitmaps are "evenly distributed" among tomography vectors, we might conclude that the average number of bitmaps per tomography vector is $\frac{2^{mn}}{B(m,n)}$. This estimate is not exact but is intended to be informative. Since $\frac{2^{750}}{B(25,30)} \approx 2 \cdot 10^{28}$, we can conclude that a "random" 25×30 digital tomography problem typically has more than 10^{28} solutions.

The following goal is to show what happens if we use slope diagonals instead of the regular ones. Let $k \in \mathbb{N}$ be a fixed slope parameter. By definition, an image contains exactly $m + \lceil \frac{n}{k} \rceil - 1$ slope diagonals (and also slope anti-diagonals). Assume that $m \leq n$ and define $I = min\{m-1, \lfloor \frac{n-1}{k} \rfloor\}$. This number represents how many slope diagonals increase in length before reaching maximum length $L = I \cdot k + min\{k, n - I \cdot k\}$. We can divide the slope diagonals into 3 categories: increase phase, plateau phase, and decrease phase. First, there are exactly I slope diagonals in the increase phase and their lengths are $k, 2k, \ldots, I \cdot k$. Next comes the plateau phase, which contains exactly P slope diagonals with length L, where

$$P = \begin{cases} m - I, & \text{if } I < m-1, \\ \lceil \frac{n}{k} \rceil - I, & \text{otherwise.} \end{cases}$$

Lastly, exactly D slope diagonals belong to the decrease phase and their lengths are $L-k, L-2k, \ldots$, where

$$D = \begin{cases} \lceil \frac{n}{k} \rceil - 1, & \text{if } I < m-1, \\ \lceil \frac{n}{k} \rceil - P, & \text{otherwise.} \end{cases}$$

Now we can express the number of possible distinct slope-diagonal sum vector as

$$\prod_{l=1}^{I} kl \cdot L^P \cdot \prod_{l=1}^{D} (L - kl)$$

For $m = 25$, $n = 30$ and $k = 2$, we can easily compute that $I = 14, L = 30, P = 11$ and $D = 14$. Using the expressions for the number of row and column sum vectors computed above, we finally obtain $B_2(25,30) \propto 10^{126}$, where $B_k(m,n)$ denotes the total number of possible combinations of row, column, and slope diagonal sums. Since $\frac{2^{750}}{B_2(25,30)} \propto 10^{99}$, we can conclude that a "random" 25×30 digital tomography problem encoded using row, column and slope diagonal sums with $k = 2$ typically has more than 10^{99} solutions.

In the last part, we substitute the diagonals with frames. As before, assume that $m \leq n$. Observe that a s-th frame f_s contains exactly $(2m + 2n - 8s + 4)$ pixels, for $s = 1, \ldots, \lceil \frac{m}{2} \rceil$. Thus, we can express the number of frame sum vectors as

$$\prod_{s=1}^{\lceil \frac{m}{2} \rceil} (2m + 2n - 8s + 4)$$

Using the number of possibilities for row and column sums computed previously, we obtain the result $B_f(25,30) \propto 10^{103}$, where $B_f(m,n)$ denotes the total number of possible combinations of row, column, and frame-based sums. Since $\frac{2^{750}}{B_f(25,30)} \propto 10^{122}$, we can conclude that a "random" 25×30 digital tomography problem encoded using row, column and frame sums typically has more than 10^{122} solutions.

3.2 Relation Between Number of Solutions and Image Size

In this section, we restrict to row, column and diagonal projections. We will show that the expected number of solutions of an $m \times n$ tomography instance increases with m and n. Note that this is an average-case statement, we cannot guarantee that for every instance larger images have more solutions.

To prove this, we will use two estimates computed in the previous section. Since each pixel has value 0 or 1, the total number of binary images is 2^{mn}. The number T of distinct tomography vectors $\leq B(m,n)$, because $B(m,n)$ counts all syntactically admissible vectors, including unrealizable ones. Recalling the reasoning mentioned above, we might deduce that the average number of solutions $= \frac{2^{mn}}{T} \geq \frac{2^{mn}}{B(m,n)}$. Showing $\frac{2^{mn}}{B(m,n)} \to \infty$ as $m, n \to \infty$ is equivalent to the desired statement.

To simplify it a little, we will use natural logarithms. Since the natural logarithm is monotone and unbounded, it is sufficient to show that $\log(\frac{2^{mn}}{B(m,n)}) = \log(2^{mn}) - \log(B(m,n)) \to \infty$ as $m, n \to \infty$. It is not difficult to show that $\log(2^{mn}) = \Theta(mn)$. However, the second term is a bit more complicated to handle.

$$\log B(m,n) = m\log(n+1) + n\log(m+1) + 2\log((m+1)!) + 2(n-m)\log(m+1) + 2\log(m!)$$

In the next step, we apply a Stirling's approximation formula: $\log k! = k\log k - k + \mathcal{O}(\log k)$.

$$\begin{aligned}
m\log(n+1) &= \Theta(m\log n) \\
n\log(m+1) &= \Theta(n\log m) \\
2\log((m+1)!) &= \Theta(m\log m) \\
2\log(m!) &= \Theta(m\log m) \\
2(n-m)\log(m+1) &= \Theta(n\log m), \text{ for } m < n
\end{aligned}$$

In the last equation, note that if $m = n$, there is only one diagonal of maximal length, and therefore this term vanishes. Thus, we can write $\log B(m,n) = \Theta(m\log n + n\log m + m\log m)$. Since $\log m \leq \log(m+n)$ and $\log n \leq \log(m+n)$, the following inequalities hold

$$\begin{aligned}
m\log n &\leq m\log(m+n) \\
n\log m &\leq n\log(m+n) \\
m\log m &\leq m\log(m+n)
\end{aligned}$$

Summing them up yields $\log B(m,n) = \mathcal{O}((m+n)\log(m+n))$. Since $mn \gg (m+n)\log(m+n)$, $\log(\frac{2^{mn}}{B(m,n)}) = \Theta(mn) - \mathcal{O}((m+n)\log(m+n)) \to \infty$ as $m, n \to \infty$. That implies the desired fact $\frac{2^{mn}}{B(m,n)} \to \infty$ as $m, n \to \infty$.

In conclusion, the number of images grows much faster than the number of distinct tomography constraints. Therefore, larger images have on average more solutions than smaller ones.

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